iii) The operators are present between the promoters and structural genes. The repressor protein binds to the operator region of the operon.

The *Lac* (Lactose) operon: The metabolism of lactose in *E.coli* requires three enzymes – permease, β-galactosidase (β-gal) and transacetylase. The enzyme permease is needed for entry of lactose into the cell, β-galactosidase brings about hydrolysis of lactose to glucose and galactose, while transacetylase transfers acetyl group from acetyl Co A to β-galactosidase.

The *lac operon* consists of one regulator gene ('i' gene refers to inhibitor) promoter sites (p), and operator site (o). Besides these, it has three structural genes namely *lac* z,y and *lac* a. The *lac* '**z**' gene codes for β-galactosidase, *lac* '**y**' gene codes for permease and '**a**' gene codes for transacetylase.

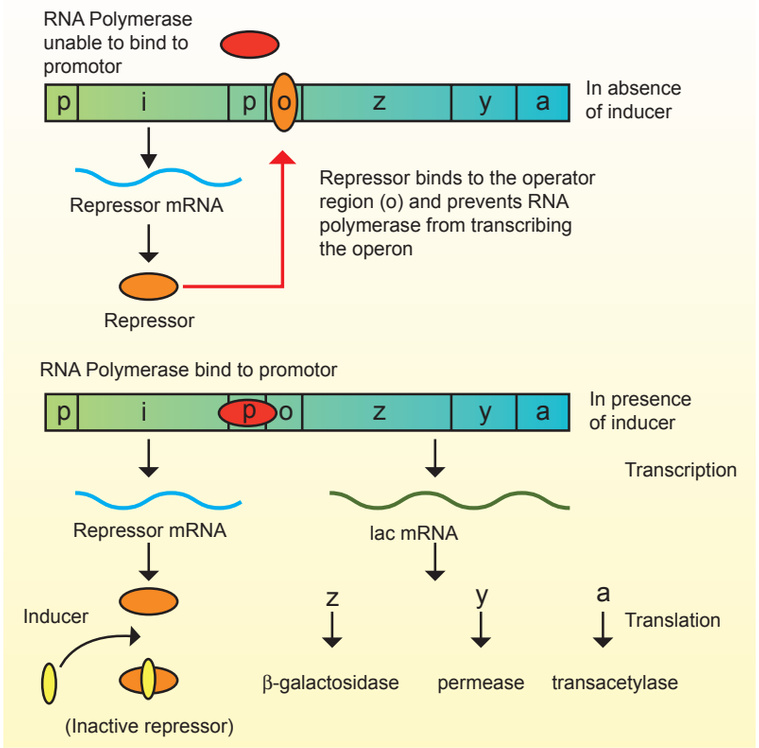


Fig. 5.14 *Lac* Operon model

Jacob and **Monod** proposed the classical model of *Lac* operon to explain gene expression and regulation in *E.coli*. In *lac* operon, a polycistronic structural gene is regulated by a common promoter and regulatory gene. When the cell is using its normal energy source as glucose, the '**i**' gene transcribes a repressor mRNA and after its translation, a **repressor protein** is produced. It binds to the operator region of the operon and prevents transcription, as a result, β-galactosidase is not produced. In the absence of glucose, if lactose is available as an energy source for the bacteria then lactose enters the cell as a result of permease enzyme. Lactose acts as an inducer and interacts with the repressor to inactivate it.

The repressor protein binds to the operator of the operon and prevents RNA polymerase from transcribing the operon. In the presence of inducer, such as lactose or allolactose, the repressor is inactivated by interaction with the inducer. This allows RNA polymerase to bind to the promotor site and transcribe the operon to produce *lac* mRNA which enables formation of all the required enzymes needed for lactose metabolism (**Fig. 5.14**). This regulation of *lac* operon by the repressor is an example of negative control of transcription initiation. *Lac* operon is also under the control of positive regulation as well.

5.14 Human Genome Project (HGP)

The international human genome project was launched in the year 1990. It was a mega project and took 13 years to complete. The human genome is about 25 times larger than the genome of any organism sequenced to date and is the first vertebrate genome to be completed. Human genome is said to have approximately 3×10^9 bp. HGP was closely associated with the rapid development of a new area in biology called bioinformatics.

5.14.1 Goals and methodologies of Human Genome Project

The main goals of Human Genome Project are as follows

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